Non-coherent receiver

A non-coherent receiver detects the transmitted signal without requiring carrier phase synchronization. Unlike coherent receivers, it does not need an exact reference carrier aligned with the transmitter.

This makes the receiver:

- ✓ Simpler
- ✓ Cheaper
- ✓ Easier to implement

But generally, with poorer BER performance.

Basic Idea: Instead of detecting the exact phase of the carrier, the receiver mainly detects:

1. Signal energy.
2. Amplitude.
3. Frequency presence.

So, it avoids complicated carrier recovery circuits.

Main components of a Non-Coherent Receiver

1. **Bandpass Filter**
   a) Receives the incoming signal
   b) Removes out-of-band noise and interference

2. **Envelope Detector / Energy Detector**

   This is the key component. It extracts:

   a) Signal envelope, OR

   b) Signal energy

   without needing phase information.

   For example: $|\boldsymbol{r(t)}|$ or $\boldsymbol{r^2(t)}$ is used to estimate signal strength.

3. Integrator / Averager

a) Measures total energy over a symbol duration
b) Helps determine which symbol was transmitted

4. Decision Device

a) Compares detected energies or amplitudes
b) Decides the transmitted bit/symbol

Example: Non-Coherent BFSK Receiver

Suppose:

a) bit 1 → frequency f_1
b) bit 0 → frequency f_2

The receiver contains:

1. two bandpass filters,
2. two envelope detectors.

Operation:

a) Each branch measures energy around one frequency
b) Larger energy determines transmitted bit

Decision:

i. Higher energy at f_1 → bit 1
ii. Higher energy at f_2 → bit 0

Why no synchronization is needed

In coherent detection: ***exact carrier phase matters.***
In non-coherent detection: ***only energy/amplitude matters.***
So even if phase changes randomly:

$$\cos(2\pi f_c t + \phi)$$

The receiver can still detect signal presence.

Advantages

1. Simple hardware
2. Lower cost
3. Easier implementation
4. No PLL or carrier recovery needed

Disadvantages

1. Higher BER
2. Lower power efficiency
3. Less accurate detection